

FUSHIKI, Japan

WMO: 476060

Lat: 36.7922N

Lon: 137.0561E

Elev: 43

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.7	29.2	20.4	15.3	36.2	22.2	16.7	35.7	19.4	39.9	17.1	39.6	5.6	230	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	92.5	77.2	89.7	77.1	87.1	76.6	79.9	87.2	78.8	86.0	77.7	84.8	8.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.9	145.0	83.9	76.7	139.4	83.0	75.6	134.1	82.1	43.5	87.4	42.3	86.0	41.2	84.8	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.9	14.4	12.4		DB	24.6	97.6	2.0	2.1	23.1	99.2	21.9	100.4	20.8	101.6	19.3	103.2
				WB	23.4	80.9	1.9	1.4	22.0	81.9	20.9	82.7	19.9	83.5	18.6	84.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.1	37.1	38.0	44.1	53.4	62.9	69.8	78.1	81.0	73.3	63.1	52.1	42.3	(6)
	DBStd	15.83	4.47	5.04	5.98	6.29	5.23	4.27	4.88	4.08	5.03	5.20	5.82	5.67	(7)
	HDD50	1264	400	339	202	35	0	0	0	0	0	0	43	245	(8)
	HDD65	3919	864	757	647	352	104	7	0	0	2	97	387	702	(9)
	CDD50	4202	0	2	20	135	400	595	872	960	698	406	107	7	(10)
	CDD65	1383	0	0	0	3	38	152	407	495	250	38	0	0	(11)
	CDH74	10626	0	0	1	39	227	535	3294	5011	1413	105	1	0	(12)
	CDH80	3423	0	0	0	7	37	105	1105	1825	327	17	0	0	(13)
(14) Wind	WSAvg	5.9	5.7	5.9	6.6	6.5	6.0	5.5	5.4	5.5	5.8	5.8	5.9	5.9	(14)
(15) Precipitation	PrecAvg	89.1	10.3	6.5	5.9	4.9	4.8	6.5	8.5	7.2	8.0	6.3	9.0	11.2	(15)
	PrecMax	107.1	17.5	11.1	8.5	9.1	12.1	12.9	20.7	18.6	15.0	10.9	17.7	24.0	(16)
	PrecMin	62.2	4.8	2.1	2.8	1.5	1.8	2.1	1.6	1.6	3.1	3.5	5.7	6.4	(17)
	PrecStd	10.9	2.8	2.2	1.7	2.0	2.6	2.6	5.1	4.1	2.9	1.8	2.4	4.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	60.8	70.5	79.3	84.3	87.6	95.6	96.1	92.2	81.9	71.9	62.1	(19)
		MCWB	46.3	51.3	53.5	58.9	65.0	72.9	77.1	77.3	75.5	68.3	59.3	52.7	(20)
	2%	DB	49.4	53.5	63.6	72.6	79.8	82.9	92.1	93.0	86.9	76.5	67.4	57.0	(21)
		MCWB	43.9	45.8	50.7	57.5	62.7	69.9	77.0	77.6	75.4	66.5	58.4	50.9	(22)
	5%	DB	46.8	49.3	58.2	68.0	76.0	79.5	89.0	90.7	83.6	73.5	64.2	53.8	(23)
		MCWB	42.1	43.7	48.7	55.2	61.5	69.4	76.7	77.8	74.2	65.1	56.8	48.5	(24)
	10%	DB	44.5	46.4	53.9	63.7	72.4	76.7	86.1	88.3	80.8	71.2	61.4	51.1	(25)
		MCWB	40.6	41.4	47.1	54.0	61.1	68.9	76.3	77.7	73.0	63.7	55.5	46.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.1	53.3	56.0	63.0	69.2	75.4	80.2	81.7	79.0	71.8	62.7	55.8	(27)
		MCDB	51.9	57.9	65.2	72.0	77.4	82.4	88.7	88.7	85.4	77.7	66.1	58.5	(28)
	2%	WB	44.8	47.4	52.7	59.7	66.1	73.3	79.0	80.5	77.0	69.3	60.5	51.8	(29)
		MCDB	48.4	52.2	60.4	68.5	73.8	78.8	86.9	87.8	83.7	73.9	65.7	56.2	(30)
	5%	WB	42.7	44.3	50.0	57.3	64.3	71.6	78.0	79.5	75.5	67.0	58.2	49.1	(31)
		MCDB	46.0	48.3	56.1	64.8	71.6	76.9	85.5	86.7	81.5	71.5	62.9	53.1	(32)
	10%	WB	40.8	41.8	47.7	55.3	62.7	70.0	77.0	78.5	74.0	64.9	56.0	46.8	(33)
		MCDB	44.2	45.9	53.3	61.7	69.7	75.0	84.2	85.6	79.5	69.9	60.7	50.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	11.1	14.5	15.3	14.5	10.9	11.0	12.0	11.4	12.9	12.8	10.7	(35)
		MCDBR	14.0	17.9	23.1	24.0	22.0	16.0	16.1	16.0	14.9	15.3	17.4	15.4	(36)
		MCWBR	9.8	12.3	13.6	12.2	10.0	7.2	5.9	6.1	6.5	7.7	10.4	10.6	(37)
	5% WB	MCDBR	13.2	16.4	19.5	18.1	16.4	12.6	13.6	13.3	12.6	12.3	14.5	14.2	(38)
		MCWBR	10.1	12.4	12.9	10.7	9.1	6.2	5.6	5.8	6.0	7.4	10.0	10.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.420	0.449	0.511	0.536	0.538	0.558	0.549	0.530	0.478	0.444	0.425	0.421	(40)
	taud		1.968	1.914	1.838	1.858	1.920	1.951	2.039	2.098	2.198	2.182	2.136	2.023	(41)
	Ebn at Noon		229	240	238	241	243	238	239	240	246	241	229	219	(42)
	Edh at Noon		43	52	62	63	61	59	54	49	43	39	37	39	(43)
(44) All-Sky Solar Radiation	RadAvg		511	772	1125	1460	1669	1588	1517	1573	1230	986	648	446	(44)
	RadStd		52	68	89	120	180	121	226	167	94	83	56	57	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	+1.08	+1.62	N/A	N/A	N/A	N/A	
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page